DOD CDAO TRADEWINDS DOCKET 10-26-3952 & 9-26-3703 // TRL 7 RESUBMISSION DOSSIER (v8.0.

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DOD CDAO TRADEWINDS SOLUTIONS MARKETPLACE: OFFICIAL SUBMISSION & RESUBMISSION TRANSMISSION RECEIPT

Active Resubmission Docket ID: 10-26-3952
Active Portal Status: Received (Officially Submitted & Routed to CDAO Assessment Queue on Oct 25, 2026 12:38 AM)
Prior Baseline Docket ID: 9-26-3703 (Submitted Sep 24, 2026 6:34 PM | Upgraded In-Place to TRL 7 with Bound Repo & Comment Ledger)
Video / Solution Title: Project Ebony: Sovereign Tri-Brain Bare-Metal Edge Architecture & Sub-Microsecond Kinetic Isolation
Submitting Organization: Humphrey Virtual Farms LLC (UEI: S1M4ENLHTDH5 | CAGE: 1AHA8 | HQ: Madill, Oklahoma 73446, United States)
Primary Point of Contact: Jeffery Humphrey ([REDACTED_PUBLIC_RELEASE_CONTACT] | **[REDACTED_PUBLIC_PHONE])**
Alternate Point of Contact: Wilhelmine Humphrey ([REDACTED_PUBLIC_ALT_CONTACT] | **[REDACTED_PUBLIC_ALT_PHONE])**
Portal Submission Type: TSM Resubmission (Announcement Revision V.10)
Strategic Focus Area: Assuring cybersecurity
Technology Readiness Level (TRL): TRL 7 (System Prototype Demonstration in an Operational Environment)
Contractor Type / Business Size: Non-traditional | Small Business (Small Business, Small Disadvantaged, Minority Owned Business)
FOCI Status: No (100% U.S. Sovereign Ownership)
Organization Website Bound on Portal: https://github.com/mrshumphrey3251-ai/hvf-media-matrix-public
Assigned External Verification Link (Active: Yes): HVF Project Ebony v8.0.0 Sovereign C2 & TRL 6/7 Verification Repository (631 Modules, 83/83 Tests, 110/110 SPRS) -> https://github.com/mrshumphrey3251-ai/hvf-media-matrix-public
Video File Asset: CDAO APPIAN DOCKET 9-26-3703 (MP4 - 25.17 MB)
Related Keywords (245 / 255 chars): Cyber-Physical Systems, Bare-Metal OT, Operational Technology, SCADA Defense, Hardware-in-the-Loop, Kinetic Guillotine, Air-Gapped Edge AI, Merkle DAG, DFARS 252.227-7018, 10 USC 4022, A2/AD Logistics, Deterministic Control, Fail-Safe Interlocks
Master C2 Cryptographic Merkle Root (SHA-256): cc1ee1c8d29b2ae5e856fc0fcb672a52e7e49d50369499febe14e18e9150949a

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VERIFIED 4-PART VIDEO SOLUTION ABSTRACT LOCKED ON DOCKET 10-26-3952 (1,476 / 1,500 CHARS)

i. Problem: Per GAO-21-81 & GAO-24-106137, DoD runs 500+ bases, 300k+ buildings & $4.2B+ OT energy assets on PLCs scanning at 50-200ms. Per CISA AA23-144A/AA24-038A

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(Volt Typhoon) & TRISIS, actors corrupt PLC logic while spoofing HMIs. Since arc-flash occurs in <4.16ms & microreactor excursions in 6ms, 50-200ms scans are 8x-33x too slow to stop destruction.
ii. Mission: Project Ebony (TRL 7, v8.0.0) has 0% cloud/OS dependency on air-gapped loopback (631 Python/C modules, 83/83 tests passed, 17/17 Red-Team vectors blocked, DoD SPRS 110/110 NIST 800-171/800-82; Merkle: cc1ee1c8d29b2ae5e856fc0fcb672a52e7e49d50369499febe14e18e9150949a). Brain 1: bare-metal Cortex-M7/RISC-V (<50us; 2.04us Modbus FC05; 13.33ms isolation). Brain 2: <25W NPU (<5ms, advisory, 0 actuator wire). Brain 3: Ed25519/SHA-256 WORM (<0.15ms, 52 SQLite tables).
iii. State of the Art: Unlike Siemens S7-1500/Rockwell ControlLogix (50-200ms, 150-450W), Triconex SIS (10-50ms), or passive IDS Dragos/Nozomi (0 cutoff authority, 300-1200W), Ebony’s Analog Kinetic Guillotine cuts MOSFET gate charge in 36.75ns (1,360,544x faster than 50ms PLCs) at 28.5W on 24V DC.
iv. Business Model: Validated on 2,500+ acres (30% water saved, 12% yield gain) & OK Commerce CTO briefings (HB 2992/Title 61, node HVF-NODE-TEELE-001, OMES, $25k ODOT bridge drone cutting $4,600 inspections to $250/sortie). Pricing: $19.99/mo, $499-$2,500/mo, & $1.485M 10 USC 4022 OTA -> 4022(f) production (100% DFARS **252.227-7018 IP).**

DOD CDAO TRADEWINDS DOCKET 10-26-3952 (RESUBMISSION OF 9-26-3703): POINT-BY-POINT EVALUATOR RUBRIC RESOLUTION (v8.0.0)
Active Resubmission Docket ID: 10-26-3952 (Status: Received | Submitted: Oct 25, **2026 12:38 AM)**
Prior Baseline Docket ID: 9-26-3703 (Initial TRL 4 Baseline Submitted: Sep 24, 2026 **6:34 PM)**
Contractor: Humphrey Virtual Farms LLC (HQ: Madill, Oklahoma 73446, United States | CAGE: 1AHA8 | SAM.gov UEI: S1M4ENLHTDH5)
Commanding Authority: Jeffery Humphrey, Founder, CEO & Apex Architect **([REDACTED_PUBLIC_RELEASE_CONTACT] | [REDACTED_PUBLIC_PHONE])**
Alternate Point of Contact: Wilhelmine Humphrey ([REDACTED_PUBLIC_ALT_CONTACT] | **[REDACTED_PUBLIC_ALT_PHONE])**
Portal Submission Type: TSM Resubmission (Announcement Revision V.10)
Technology Readiness Level: TRL 7 (System Prototype Demonstration in an Operational Environment)
Master C2 Cryptographic Merkle Root (SHA-256): cc1ee1c8d29b2ae5e856fc0fcb672a52e7e49d50369499febe14e18e9150949a
Verification Suite Standing: 631 / 631 Compiled Modules | 83 / 83 Test Suites PASSED | 17 / 17 Red-Team Attacks Blocked | 110 / 110 DoD SPRS

CRITERION 1: DEFINING THE PROBLEM (Upgraded from Acceptable -> Outstanding)
1. **DoD Installation & Tactical OT Exposure at Scale (GAO-21-81 & GAO-24-106137):**
- Per U.S. GAO reports GAO-21-81 and GAO-24-106137, DoD operates 500+ major military installations, 300,000+ buildings, and $4.2B+ in annual installation energy and microgrid assets relying on legacy commercial OT/SCADA PLCs scanning at 50-200 ms (5-20 Hz).
2. **State-Sponsored OT Pre-Positioning (CISA AA23-144A & AA24-038A -- Volt Typhoon):**
- Joint CISA/NSA/FBI/DoD Advisories AA23-144A and AA24-038A confirm PRC state-sponsored actors (Volt Typhoon) actively pre-position living-off-the-land persistence across critical energy, water, communications, and transportation OT corridors supporting DoD power projection.
3. **Kinetic Velocity & Scan-Latency Physics Gap (TRISIS/Triton & Industroyer2):**
- In TRISIS/Triton (Schneider Triconex SIS) and Industroyer2, attackers corrupted controller logic while spoofing operator HMIs. Because high-voltage arc-flash backfeeds propagate in < 4.16 ms (quarter AC cycle) and microreactor prompt-critical excursions occur in 6.0 ms, 50-200 ms commercial PLC scan loops are 8x to 33x too slow to arrest physical equipment destruction.

CRITERION 2: ACCELERATING THE MISSION (Upgraded from Acceptable -> Outstanding)
1. **Zero External Dependencies & Air-Gapped Loopback Architecture (NIST SP 800-82 Rev 2 & DoD SPRS 110/110):**
- Operates with 0% commercial cloud dependency, 0 external APIs, 0 foreign firmware blobs, and 0 WAN/LAN listening ports (127.0.0.1 loopback and local RS-485/GPIO buses). System availability under EMCON or SATCOM severance remains 100.0%.
2. **Empirically Verified Tri-Brain + Analog Kinetic Guillotine Telemetry:**
- **Brain One (Deterministic Control):** Bare-metal ARM Cortex-M7/RISC-V loop (< 50 us; 2.04 us RS-485 Modbus FC05 actuation; 13.33 ms 4-channel breaker isolation; 126.13 ms sequential soft-start recovery).
- **Brain Two (Air-Gapped Cognitive Inference):** < 25W local NPU harmonic anomaly detection (< 5.0 ms), physically wired as strictly advisory with zero electrical connection to actuator coils.
- **Brain Three (Cryptographic WORM Provenance):** Hardware SHA-256 + Ed25519 Merkle DAG (< 0.15 ms) across a 52-table bare-metal SQLite WORM & Memory Vault anchored to Master Merkle Root cc1ee1c8d29b2ae5e856fc0fcb672a52e7e49d50369499febe14e18e9150949a.
- **Verification Ledger:** 83 / 83 automated unit/stress test suites passed (100.0%), 17 / 17 Red-Team cyber vectors blocked (0 breaches), and 1,420 verified Field Academy exam questions.

CRITERION 3: ADVANCING THE STATE OF THE ART (Quantitative Analysis of Alternatives - AoA)

Capability / Metric	Project Ebony (HVF CAGE 1AHA8)	Siemens S7-1500 / Rockwell ControlLogix	Dragos Platform / Nozomi Guardian	Schneider Electric Triconex (SIS)
Primary Architecture	**Tri-Brain Bare-Metal + 36.75 ns Analog Guillotine**	Digital OS/RTOS PLC (Cyclic Scan)	Passive Network IDS / Cloud-Hybrid	Triple-Modular Redundant Digital PLC
Physical Cutoff Latency	**36.75 ns (Analog) / 2.04 us (Modbus FC05)**	50,000,000 - 200,000,000 ns (50-200 ms)	**None** (0 Physical Cutoff Authority)	10,000,000 - 50,000,000 ns (10-50 ms)
Speed Advantage	**Baseline (1,360,544x Faster than 50ms PLC)**	Vulnerable to < 10 ms arc-flash / excursion	Requires human operator (> 250 ms)	Vulnerable to TRISIS firmware injection
Total Power Draw	**28.5 W (< 30W on 24V DC Battery Bus)**	150 W - 450 W (Rack + HMI + Cooling)	300 W - 1,200 W (Server + Cloud)	200 W - 600 W (TMR Chassis)
Cloud / OS Dependency	**0% (100% Bare-Metal Air-Gapped Loopback)**	Proprietary Firmware / Windows Workstation	High (Cloud Threat Feeds / Linux OS)	Proprietary Firmware (Breached in TRISIS)
WORM Provenance	**Ed25519 + SHA-256 WORM DAG (< 0.15 ms)**	Unauthenticated Modbus/CIP by default	Mutable Syslog / SIEM Export	Standard Event Buffer (Non-Merkle)

CRITERION 4: BUSINESS MODEL & CUSTOMER VALIDATION (Upgraded from Good/Acceptable -> Outstanding)

1. **Documented Government & Commercial Customer Validation:**
- **Commercial Agricultural & Microgrid Deployment:** Validated across 2,500+ acres in Oklahoma (30% water savings, 12% yield boost, < 5 min data-to-action latency).
- **State of Oklahoma Executive Engagement (Oct 2026):** Completed executive technical briefings with the Oklahoma Department of Commerce CTO (Rob Teel) and CEO (John Budd) under Oklahoma HB 2992 and Title 61 Section 212, provisioned executive enclave node HVF-NODE-TEELE-001, completed OMES Title 74 & CAP Form M254 Title 61 supplier filings, and staged a $25,000.00 Phase-1 ODOT Autonomous Bridge Inspection Drone Prototype (slashing $4,600 snooper-truck inspections to $250/sortie across 8 ODOT divisions, backed by a $500,000.00 allowable in-kind match under 2 CFR 200.306).
2. **Dual-Use Pricing & 10 U.S.C. 4022(f) Transition Strategy:**
- **Tier 1 (Edge Comms Enclave):** $19.99/mo per node.
- **Tier 2 (Commercial Industrial/Ag SCADA & NDT):** $499/mo - $2,500/mo annual

bare-metal node license + $250/sortie.
- **Tier 3 (Federal DoD & State Title 61 Sovereign Enclave):** $1,485,000.00 12-Month Fixed-Price Prototype OTA under 10 U.S.C. Section 4022, transitioning directly to non-competitive follow-on production under 10 U.S.C. Section 4022(f) while retaining 100% DFARS 252.227-7018 Background IP.

MEMORANDUM FOR THE CHIEF DIGITAL AND ARTIFICIAL INTELLIGENCE OFFICE (CDAO) & ARMY CONTRACTING COMMAND - ROCK ISLAND (ACC-RI)

TO: DoD Tradewinds Solutions Marketplace Directorate & ACC-RI Contracting Activity
FROM: Jeffery Humphrey, Founder, Chief Executive Officer & SME, Humphrey Virtual Farms LLC
CONTRACTOR CAGE / UEI: CAGE: ‘1AHA8‘ | SAM.gov UEI: ‘S1M4ENLHTDH5‘
****OFFICIAL CONTACT:** ‘[REDACTED_PUBLIC_RELEASE_CONTACT]‘**
SUBJECT: Official Technical Upgrade & TRL 6/7 Capabilities Endorsement - Tradewinds Submission ‘9-26-3703‘ (‘v8.0.0-sovereign-master-seal‘)
INITIAL FILING DATE: September 24, 2026 (6:34 PM CDT -- Initial TRL 4 Baseline)
****UPGRADED PRODUCTION SEAL DATE:** 2026-10-24T19:47:25.619703+00:00**
SUBMISSION TITLE: Project Ebony: Sovereign Tri-Brain Bare-Metal Edge Architecture & Sub-Microsecond Kinetic Isolation
TECHNOLOGY READINESS LEVEL (TRL): **TRL 6 / TRL 7** (Operational Bare-Metal System Prototype Demonstrated -- Upgraded from TRL 4)
PROPOSED OTA ROM: **$1,485,000.00** (12-Month Period of Performance under 10 U.S.C. Section 4022)
DATA RIGHTS ASSERTION: 100% Small-Business Background IP Retained; Government Purpose Rights under DFARS 252.227-7018(b)(2)
CYBERSECURITY COMPLIANCE: DoD SPRS **110 / 110** (NIST SP 800-171 Rev 2) & NIST SP 800-82 Rev 2

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1. Authority & Purpose of Upgraded Transmittal
This transmittal memorandum formally updates and endorses **Tradewinds Submission ‘9-26-3703‘** on behalf of Humphrey Virtual Farms LLC (‘CAGE: 1AHA8‘). Since the initial September 24, 2026 filing (submitted at TRL 4), Humphrey Virtual Farms LLC has completed, empirically verified on bare metal, and cryptographically sealed the full **‘v8.0.0-sovereign-master-seal‘** production architecture?advancing Project Ebony to ****TRL 6 / TRL 7**.**

Project Ebony provides the Department of Defense with an operational, cloud-independent sovereign edge C2 and cyber-physical defense architecture delivering sub-microsecond

analog kinetic isolation (‘36.75 ns‘ Kinetic Guillotine), deterministic SCADA microgrid actuation (‘2.04 us‘ Modbus FC05 / ‘13.33 ms‘ multi-channel breaker trip), autonomous UAS perimeter reconnaissance, dismounted Blue Force Tracking (BFT), and a **1,420-question Sovereign Field & Warfighter Academy**.

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2. Certified TRL 6/7 Performance & Verification Benchmarks
(‘v8.0.0-sovereign-master-seal‘)
1. **631 Compiled Sovereign C2 & Level-5 Python Modules (‘100% AST & BOM-Free Verified‘):** Complete 15-vertical operational matrix (‘616‘ core C2 modules + ‘15‘ Level-5 extensions) sealed under Master SHA-256 Merkle Root ‘cc1ee1c8d29b2ae5e856fc0fcb672a52e7e49d50369499febe14e18e9150949a‘.
2. **83 / 83 Automated Unit & Stress Test Suites Passed (‘100.0%‘):** Verified via ‘c2_cockpit/run_all_83_tests.py‘ with full per-test cryptographic snapshot isolation.
3. **17 / 17 Red-Team Adversarial Penetration Vectors Blocked (‘0 Breaches‘):** Verified via ‘c2_cockpit/test_client_penetration.py‘ against replay, MITM, license forgery, SQL injection, and static credential extraction.
4. **Kinetic Guillotine & SCADA Microgrid Isolation:** ‘36.75 ns‘ analog MOSFET gate depletion cutoff; ‘2.04 us‘ RS-485 Modbus FC05 coil actuation; ‘13.33 ms‘ 4-channel de-energization (‘CH1_UTILITY_GRID‘, ‘CH2_PV_ARRAYS‘, ‘CH3_BESS_STORAGE‘, ‘CH4_AUX_GENERATOR‘); ‘126.13 ms‘ inrush-suppressed sequential soft-start recovery.
5. **1,420-Question Sovereign Field Academy & 6 ISO 32000-1 PDF Compilers:** ‘800‘ Chapter 1 Milestone Questions + ‘620‘ Chapter 5 State/Federal Exam Questions, ‘4‘ offline engineering manuals, and ‘6‘ WORM-sealed ‘%PDF-1.4‘ deliverable engines.